Design Document -

Airport Investment Intelligence Agent

This AI agent helps investment analysts identify profitable U.S. airport expansion opportunities through natural language chat, so users can ask questions like "Which airports in New England need terminal expansion?" and receive data-driven recommendations. It covers about 60 major U.S. airports, analyzing congestion, growth trends, and expansion options within a single conversational workflow that streamlines data retrieval and scoring.

A critical design principle is that the LLM determines which tools to call and explains the results, while all scores are computed deterministically in Python using auditable formulas, with no hallucinated statistics.

[Github repo]

System Architecture

The system uses a three-layer architecture with strict separation of concerns to keep the user experience responsive while ensuring every numeric output is reproducible.

This structure makes it easy to validate outputs, trace decisions, and maintain confidence in the recommendations.

Layer	**Technology**	**Responsibility**
Chat UI	Vite + React	Sends messages to backend, renders markdown responses, voice input/output
Agent Backend	Python + FastAPI	LLM orchestrator using Claude with tool-calling pattern
Scoring Engine	Pure Python	Deterministic formulas, no LLM in number crunching

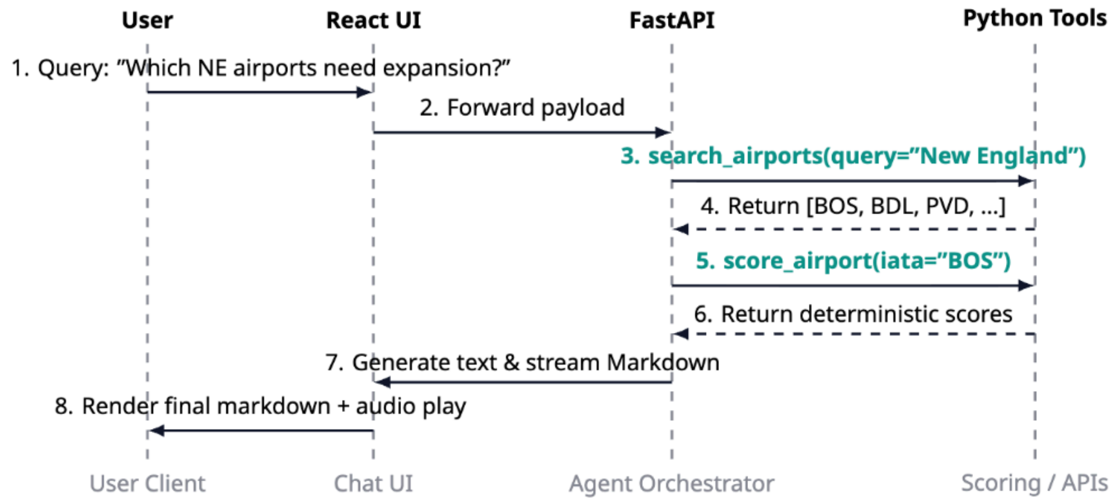
Available Tools

The agent has access to 6 deterministic tools for data retrieval and scoring.

Tool	Parameters	Data Source	Purpose
search_airports	query (string)	Static dataset	Find airports by name, city, state, or region
get_airport_stats	iata (string)	Static dataset	Retrieve passenger counts, delays, load factors
score_airport	iata (string)	Scoring engine	Compute all investment scores (congestion, growth, capacity, long-haul)
compare_airports	iata[] (array)	Scoring engine	Generate side-by-side comparison of multiple airports
get_flight_distribution	iata (string)	Static dataset	Get short/medium/long-haul flight breakdown
get_airport_status	iata (string)	FAA API (live)	Retrieve real-time delays, weather, ground stops

LLM Tool-Call Flow

The agent follows a repeatable loop: it receives the user's question, the LLM decides which tools to call, the tools execute against deterministic datasets, and the results are fed back into the model. This cycle continues until the LLM has enough information to produce a final answer with clear reasoning and caveats. To prevent runaway behavior or infinite loops, the workflow is capped at 10 iterations.



AI Responsibility Matrix

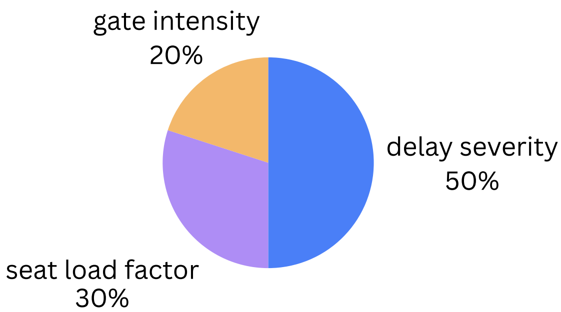
Responsibility	Owner	Rationale
Understand user intent	**LLM (Claude)**	Natural language parsing and query interpretation
Plan tool-call sequence	**LLM (Claude)**	Multi-step reasoning and decision-making
Fetch airport data	**Tools (Python)**	Deterministic lookup from static datasets
Compute investment scores	**Tools (Python)**	Fixed formulas, fully auditable, no hallucination risk
Explain results	**LLM (Claude)**	Interpret scores, provide context, flag caveats and limitations

Scoring Methodology

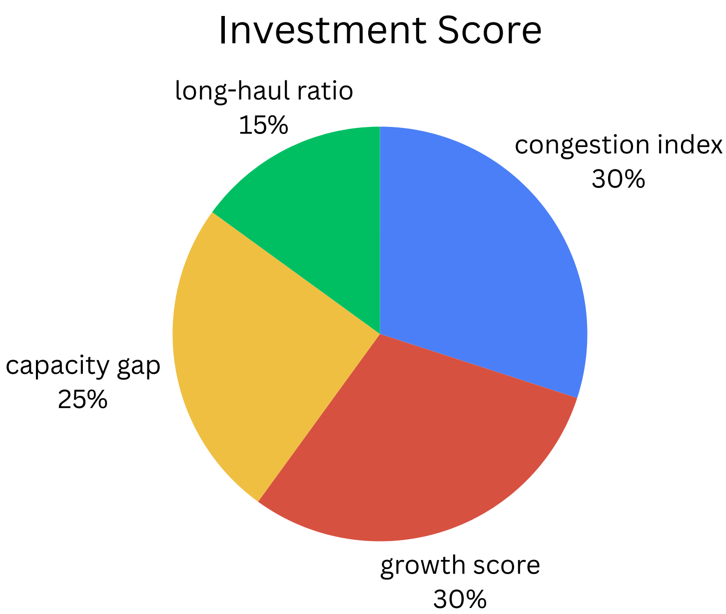
All scores range 0–100 and are computed via deterministic Python formulas. Four component scores feed into a weighted composite Investment Score.

1. **Congestion Index** - How overcrowded is the airport? This index is divided into three key aspects:

- Gate Intensity - How long are flights sitting on the ground past their scheduled departure? Higher average minutes = more points.
- Seat Load Factor - How full are the planes?
- Delay Severity - How hard is each gate working?



2. **Growth Score** - How fast is the airport gaining passengers?
3. **Capacity Gap** - How close is the airport to running out of physical space?
4. **Long-Haul Ratio** - How much of the airport's traffic is long-distance?
5. **Investment Score** - Weighted combination of all factors.



Assumptions & Limitations

The system is transparent about its constraints and communicates them in every response.

- Gate counts are estimates from public sources, not official airport data
- Capacity formula uses ~5.5 flights/gate/day industry average (may vary by airport)
- Coverage limited to ~60 major US airports only
- Historical data from 2019 + 2022–2024 BTS (COVID years 2020-2021 excluded)
- Scoring weights (30/30/25/15) are opinionated, not empirically validated
- No consideration of regulatory, environmental, or political factors
- Agent explicitly communicates these limitations in every response to users

Data Sources

Source	**Type**	**Description**
BTS - Bureau of Transportation Statistics	Static	Passenger counts, delays, load factors from 2019, 2022-2024 for ~60 airports
FAA Airport Status API	Live	Real-time delays, ground stops, weather conditions (no API key required)